

RAG

Retrieval Augmented

Generation

What is RAG and what problems does it solve?

What is the architecture and how do you set up the project?

What kind of data will be used in this project? Terminology?

AI Query: Prompt + Context

System Prompt:

You are a helpful assistant. Use the context below to answer the question. If the context does not contain the answer, say: "I could not find that information."

Context:

<Company Sick Leave Standard Operating Procedures>

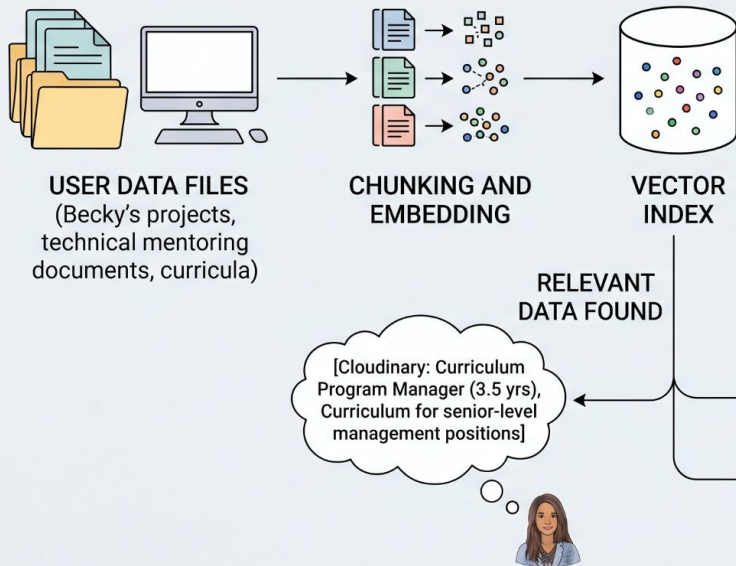
User Query:

What can I use sick leave for?

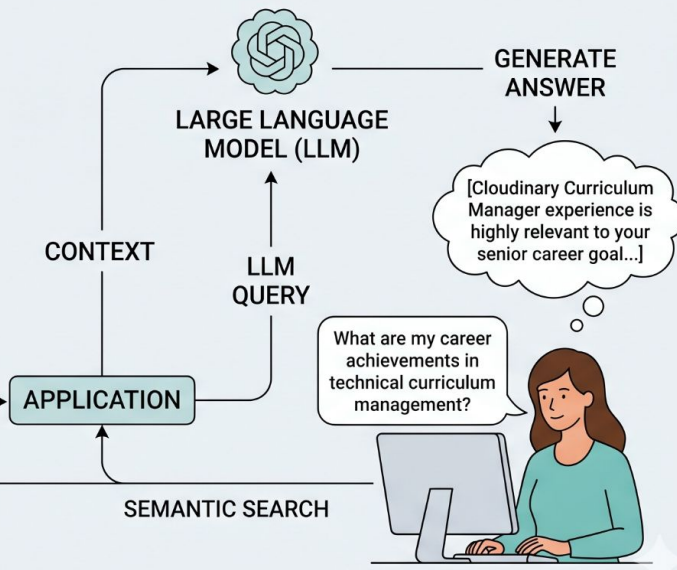
Visual Overview of RAG

UNDERSTANDING RAG: RETRIEVAL-AUGMENTED GENERATION

STAGE 1: DATA INGESTION



STAGE 2: RETRIEVAL AND GENERATION



What are the problems that RAG solves?

1

User Context

Make user data available as Context for a locally managed Large Language Model.

2

Security

By providing user data to your LLM locally, you can avoid many security issues.

3

Lower Cost

Save money on tokens and LLM cloud based plans by using a local LLM.

4

Reduce Hallucinations

Hallucinations are minimized when LLM data has access to up to date user data.

Benefits to Using RAG

1

ACCURACY

2

PRIVACY

3

NO TOKEN COST

4

**REDUCED
HALLUCINATIONS**

RAG's 2 Phase Pipeline Architecture

Python Code and Packages



Local Data Ingestion and Indexing

1. Convert raw data input to text.
2. Convert text to numerical vectors.
3. Create a searchable index using the numerical vectors.



Data Retrieval and Inference

1. User enters query and prompt is created.
2. Data is retrieved from use data index.
3. Comparison between query text and user data is enhanced by ranking similarity.
4. Highly ranked user data is passed to LLM as context.

Docker Model Runner

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AI AND AGENTS

Overview

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Inference engines

Gordon

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Docker Model Runner

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Requires: Docker Engine or Docker Desktop (Windows) 4.41+ or Docker Desktop (MacOS) 4.40+



For: See Requirements section below

Docker Model Runner (DMR) makes it easy to manage, run, and deploy AI models using Docker. Designed for developers, Docker Model Runner streamlines the process of pulling, running, and serving large language models (LLMs) and other AI models directly from Docker Hub, any OCI-compliant registry, or [Hugging Face](#).

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How Docker Model Runner works

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docker model is not recognised

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Sample User Data for the Course

- Employee Onboarding SOP.pdf
- Healthcare Benefits SOP.pdf
- Recognition and Rewards SOP.pdf
- Sick Leave Policy SOP.pdf

Terminology

RAG	Retrieval Augmented Generation refers to adding updated user data to query context
LLM or “Model”	Model is a general form of a program analyzes data and forms patterns, while an LLM is a specific type of Model focused on human language
Fine Tuning	Training an existing AI Model to perform tasks by exposing it to additional data instead of retraining it
Grounding	Anchoring an LLM response to specific data retrieved from a vector database created with up to date user data
Docker Model Runner	Set of tools to manage and run AI models locally
Data Conversion	Process of converting user data to a text format that can be converted into numeric vectors
Ingestion Pipeline	Set of steps needed to convert user data into a vector database
Query Pipeline	Set of steps needed to input user query and retrieve relevant user data to provide an answer

How do I set up the project to run locally?

- **See videos for Section 1**
 - **Setup: Create Virtual Environment using Python version 3.13**
 - **Setup: Docker Model Runner**

RAG Retrieval Augmented Generation

At this point, you're ready to run the demo lectures in Section 1.
These slides will be available as a PDF for review.
